ST308 Project: Do Past Prices Predict Future Returns? A Bayesian Test of the Efficient Market Hypothesis Using Dow Jones Index Stocks

Candidate Number: 60558

April 15, 2026

Contents

1 Introduction **3**

2 Data Description **3**

3 Methodology **5**
3.1 Bayesian Inference and MCMC 5
3.2 Task 1 - Bayesian Linear Regression 5
3.3 Task 2 - Bayesian Hierarchical Linear Model 6

4 Results: Bayesian Linear Regression **7**
4.1 Model Comparison 7
4.2 Posterior Estimates 7
4.3 Interpretation 7

5 Results: Hierarchical Model **7**
5.1 Sector-Level Heterogeneity 7
5.2 Interpretation 9

6 Conclusion **9**

7 Limitations **10**

1 Introduction

One of the most enduring ideas in financial economics is the Efficient Market Hypothesis (EMH), proposed by Fama (1970). In this project, weak form tests are tested in which the information set is just the historical price as asserted by the EMH. This means that no trading strategy based solely on historical data will consistently generate above average returns. Past prices cannot predict future ones.

However, many investors have tried to use various technical indicators to predict future returns of the stock market. Various academic literature on technical analysis have tested this hypothesis of whether these signals truly have predictive content. This is a crucial question to discuss, as even a simple and reliable signal has direct implications of market efficiency.

This project tests the weak form EMH using weekly trading data of 30 stocks from the Dow Jones Industrial Average (DJIA) over the first two quarters of 2011, drawn from the UCI Machine Learning Repository (Brown, 2013). The 2 main questions addressed are:

1. Do technical indicators derived from historical prices such as weekly price change, volume change, dividend proximity contain statistically meaningful information about the next week's return?

2. Does any such predictive relationship vary across industry sectors, or is it uniform across the market?

We use the Bayesian Markov Chain Monte Carlo (MCMC) methods to address both questions. For the first question, we fit the Bayesian linear regression of next week's percentage return on a set of technical indicators. To address the second problem, a Bayesian hierarchical model is fit into weekly returns nested within industry sectors to explore how the strength of predictive relationships vary between sectors.

2 Data Description

The dataset contains 750 observations across 30 DJIA stocks over January-June 2011. Each row corresponds to a single stock for a single week. The 30 stocks span a wide range of industries, from financial services (JPMorgan Chase, Bank of America, American Express, Travelers Companies) to technology (IBM, Intel, Microsoft, Cisco HP Inc.), energy (ExxonMobil, Chevron), healthcare (Johnson & Johnson, Merck, Pfizer), consumer goods (Coca-Cola, McDonald's, Kraft, Procter & Gamble, Walmart, Walt Disney), industrials (Boeing, Caterpillar, 3M, General Electric, United Technologies), telecoms (AT&T, Verizon), materials (DuPont, Alcoa), and Consumer Discretionary (Home Depot).

We assign each stock to one of six industry sectors based on the standard Global Industry Classification Standard (GICS) classification. This is used to address

the second problem, where we use the Bayesian hierarchical model.

Sector	Stocks (n)
Financials	BAC, AXP, JPM, TRV (4)
Technology	IBM, INTC, MSFT, CSCO, HPQ (5)
Energy	XOM, CVX (2)
Healthcare	JNJ, MRK, PFE (3)
Consumer	KO, MCD, KRFT, PG, WMT, DIS (6)
Industrials/Other	BA, CAT, MMM, GE, UTX, DD, HD, AA, T, VZ (10)

Table 1: Stock Classification

The response variable is the percentage change in the stock's price in the following week. Its distribution across all observations is approximately symmetric around zero (mean = 0.19 % , SD = 2.66 %). The variable also has a kurtosis of 5.31 and is negatively skewed, indicating a heavier left tail. Hence the handful of observations that sit clearly outside where the normal distributions have reached zero at both tails is consistent with the documented fat-tailed fact of equity returns (Fama, 1965).

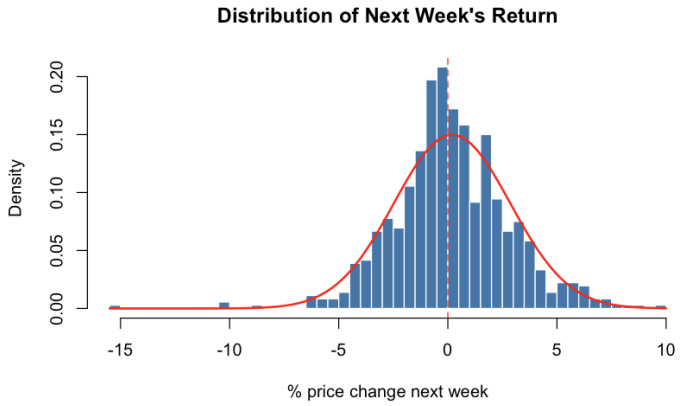


Figure 1: Distribution of Next Week's Return Overlaid with Normal Distribution

The predictors used as technical indicators are the following:

- Weekly price change
- Weekly volume change
- Proximity to the next dividend payment
- Expected dividend yield

All continuous predictors are standardised prior to modelling to facilitate prior specification and coefficient comparison later.

3 Methodology

3.1 Bayesian Inference and MCMC

All models are estimated within a Bayesian framework using Markov Chain Monte Carlo (MCMC) sampling via the 'rstanarm' package in R. Rather than producing single point estimates of model parameters, Bayesian inference yields posterior distributions where the values of each model parameters draw from given the data and prior information. This tests the EMH so that instead of a binary hypothesis test, we can observe if the posterior probability mass lies near zero (consistent with efficiency) or is away from zero (consistent with predictability).

We use the default weakly informative priors provided by 'rstanarm' as recommended by Gelman et al. (2013). 'rstanarm' automatically scales the normal priors to the specific data, applying a normal prior on regression coefficients with adjusted scale 6.7 (Figure 2) and a normal prior on the intercept centred at 0.19 with the same scale.

```
Priors for model 'bmod1'
------
Intercept (after predictors centered)
  Specified prior:
    ~ normal(location = 0.19, scale = 2.5)
  Adjusted prior:
    ~ normal(location = 0.19, scale = 6.7)

Coefficients
  Specified prior:
    ~ normal(location = 0, scale = 2.5)
  Adjusted prior:
    ~ normal(location = 0, scale = 6.7)
```

Figure 2: Scale and Mean Applied by 'rstanarm'

We then run 4 parallel MCMC chains of 2000 iterations (1000 warm-up), which yields 4000 posterior draws.

3.2 Task 1 - Bayesian Linear Regression

The response variable 'percentage change next weeks price' is continuous hence linear regression is used. The model is as follows:

$$Y_{it} = \beta_0 + \beta_1 \cdot \Delta P_{it} + \beta_2 \cdot \Delta V_{it} + \beta_3 \cdot D_{it} + \beta_4 \cdot R_{it} + \varepsilon_{it}$$

where Y_{it} is next week's return for stock i at time t; ΔP_{it} is this week's price change; ΔV_{it} is volume change; D_{it} is days to next dividend; R_{it} is percent return on next dividend; and $\varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$.

We fit two competing models: a null model with momentum and intercept only, and the full technical indicator model with four predictors. Model comparison uses Leave-One-Out cross-validation (LOO-CV) via the 'loo' package (Vehtari, Gelman and Gabry, 2017), which estimates out-of-sample predictive accuracy. According to Gelman et al. (2013), LOO-CV is used instead of other measurements for the following reasons. BIC is not intended to predict out-of-sample model performance. AIC and DIC estimate the performance of the plug-in predictive density. WAIC is asymptotically equivalent to LOO-CV, but in our limited dataset of 24 observations per stock, LOO-CV is more reliable.

3.3 Task 2 - Bayesian Hierarchical Linear Model

To address the second task, we seek to investigate whether heterogeneity exists between sectors: technology stocks may exhibit momentum effects that financial stocks do not and energy stocks may respond to commodity price signals. Hence, we now extend to a two-level hierarchical model:

$$Y_{ij} = (\beta_0 + u_{0j}) + (\beta_1 + u_{1j}) \cdot \Delta P_{ij} + \beta_2 \cdot \Delta V_{ij} + \beta_3 \cdot D_{ij} + \beta_4 \cdot R_{ij} + \varepsilon_{ij}$$

where $j \in \{1, \ldots, 6\}$ indexes sector, and the random effects follow:

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0},\ \mathbf{\Sigma}\right)$$

The random intercept u_{0j} captures sector-level differences in average return; the random slope u_{1j} allows the momentum effect (ΔP) to vary by sector. The variance components in $\mathbf{\Sigma}$ are the key quantities of interest: a large τ_1^2 (variance of slopes) would indicate that momentum has meaningfully different predictive content across sectors, a violation of market-wide efficiency even if the average effect is small.

Random slopes are included only for momentum, which is the predictor most theoretically relevant to the weak-form EMH test. The posterior coefficient for volume change is effectively zero (Table 2). Dividend date and dividend yield are publicly disclosed information and hence belong to the semi-strong form of EMH. Any movement caused by this information is a rational, mechanical adjustment, hence we do not include a sector specific intercept for these variables.

The model is estimated using "stan_glmer()" in 'rstanarm' package.

4 Results: Bayesian Linear Regression

4.1 Model Comparison

LOO-CV comparison shows a negligible difference between the momentum-only model and the full technical indicator model (ELPD difference = -1.4, SE = 3.1). Since the difference is smaller than the standard error, there is no evidence that adding volume change, dividend proximity and dividend yield improves out-of sample predictive accuracy beyond momentum alone. Hence this is generally consistent with the weak-form EMH.

4.2 Posterior Estimates

Predictor	Posterior Median	95% Credible Interval
Intercept	0.2	(-0.003, 0.390)
Momentum	0.1	(-0.146, 0.259)
Volume change	0.0	(-0.154, 0.244)
Days to dividend	-0.1	(-0.303, 0.091)
Dividend yield	0.3	(0.084, 0.489)
σ	2.7	(2.522, 2.801)

Table 2: Posterior Estimates, Bayesian Linear Regression of next-week Return

4.3 Interpretation

The most notable result is the dividend yield. It is the strongest and most credible predictor. The entire CI is positive, indicating strong evidence that higher dividend yield predicts higher next-week returns. For momentum, there is a positive median but the CI includes zero. Hence this is a weak evidence that momentum may matter but we cannot rule out that it has no effect on next week's price change. No effect is caused by volume change, and a negative median from dividend proximity suggests that returns may be lower further from the dividend date.

$\sigma = 2.7$ is quite large relative to the coefficients, indicating that the predictors explain very little of the week-to-week variance.

5 Results: Hierarchical Model

5.1 Sector-Level Heterogeneity

The hierarchical model allows each sector to have its own baseline return level and its own momentum coefficient. From Figure 3, the graphs are all centred around 0.19% to 0.20%. We observe that Industrials has the tallest peak, and this may be due to the fact that it is the largest sector group. Whereas from

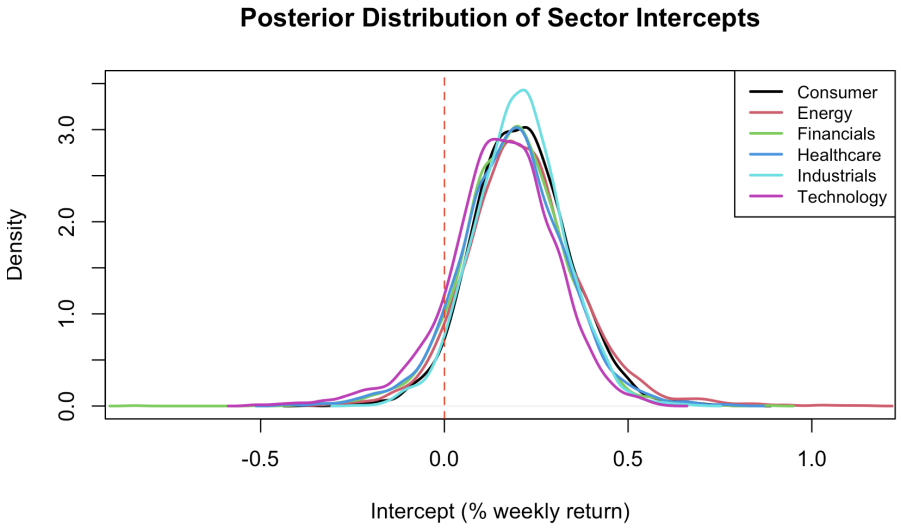


Figure 3: Posterior Distribution of Sector Intercepts

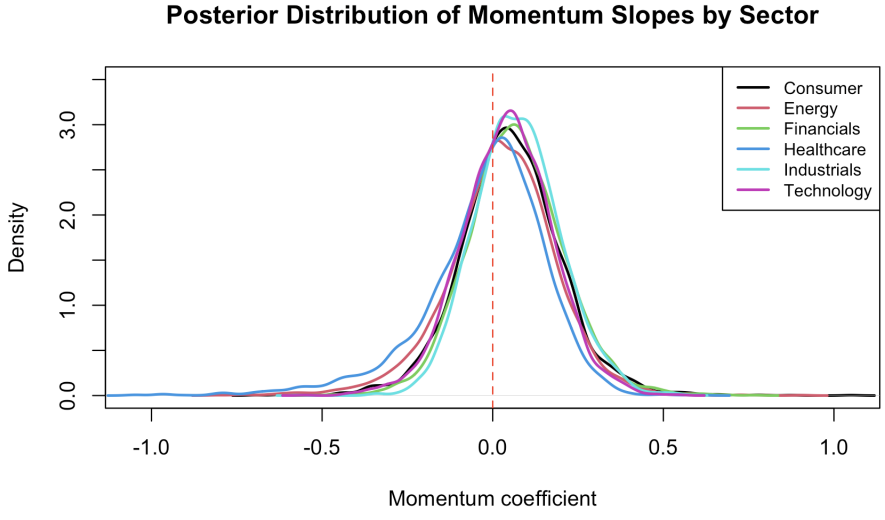


Figure 4: Posterior Distribution of Momentum Slopes by Sector

Figure 4 we observe that the posterior distributions were centred around 0.04% to 0.05%. This solidifies the EMH conclusion, where the posterior distributions of the intercepts and momentum slopes of the 6 different sectors are so similar that they almost overlap each other.

The estimated between sector variance in intercept only is $\hat{\tau}_0^2 \approx 0.013$ and the posterior median (within sector) is $\sigma^2 \approx 7.0477$. The proportion of total variance contributed by sector membership is only:

$$\frac{\hat{\tau}_0^2}{\hat{\tau}_0^2 + \hat{\sigma}^2} = \frac{0.013}{0.013 + 7.05} = \frac{0.013}{7.063} \approx 0.18\%$$

This is a very strong result in favour of market efficacy. The hierarchical model finds that knowing which sector a stock belongs to tells us almost nothing about its weekly return - sectors behave nearly identically in terms of baseline return levels. The variation that matters is happening at the individual stock and week level, not at the sector level.

The sector-specific momentum slopes are all very close to zero so there is no evidence that momentum predicts returns differently across sectors.

5.2 Interpretation

The hierarchical model uses partial pooling: this is where each sector's estimates are pulled toward the overall mean , and this is inversely proportional to sample size. In this case, the Energy sector (n = 2 stocks) will be shrunk most aggressively; the Industrial/Other sector (n = 10 stocks) will be the most data-driven.

In this section, we have gained clarity on the fact that no technical indicator works across the market as whole, and this applies to within specific sectors as well, since we have only obtained a small contribution from the between sector variance out of the overall variance for the intercept. Each slope momentum is also close to zero as seen in Figure 4.

6 Conclusion

This project has tested the weak-form Efficient Market Hypothesis using a Bayesian hierarchical framework to 30 DJIA stocks in the first two quarters of 2011. Two models were tested via MCMC - the Bayesian linear regression model and a two-level hierarchical model.

The results of both models support the EMH. The Bayesian linear regression yielded a high variance in comparison to near zero coefficients, and the hierarchical model had a non-zero but extremely small between sector variance $\hat{\tau}_0^2$, which is not practically exploitable anyway.

The Bayesian framework tells us that there is a near-zero predictability for next-week's returns, and quantifies how near each coefficient is to zero. This may be

more useful information compared to a frequentist linear regression, where we debate whether to reject the null hypothesis entirely.

7 Limitations

The dataset covers only 2 quarters of 2011, which is specific to a post-2008 recovery period and the European sovereign debt crisis. Hence this specific macroeconomic environment limits generalisability. With 750 observations across 30 stocks, the panel is short. There is also only four technical indicators in this dataset, indicators such as moving average crossovers could also be constructed from raw price data and included for future work.

[3] [1] [2] [5] [4]

References

[1] Brown, M. (2013). Dow jones index. UCI Machine Learning Repository.

[2] Fama, E. F. (1965). The behavior of stock-market prices. *The Journal of Business*, 38(1):34–105.

[3] Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417.

[4] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. CRC Press, 3 edition.

[5] Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5):1413–1432.